EXPLANATIONS

1. (*a*) B_1 = initial magnetic field = $\mu_0 ni$

B_2 = final magnetic field = μni

% change in $B = \dfrac{B_2 - B_1}{B_1} \times 100\%$

$= \dfrac{\mu ni - \mu_0 ni}{\mu_0 ni} \times 100\% = \dfrac{(\mu - \mu_0)}{\mu_0} \times 100\%$

We know $\mu = \mu_0 \mu_r$

$= \dfrac{(\mu_0 \mu_r - \mu_0)}{\mu_0} \times 100\%$

$= (\mu_r - 1) \times 100\%$

$= \chi_n \times 100\%$, $= 1.2 \times 10^{-3}\%$

2. [1] The magnetic field from straight segments AB and CD is zero. The net field B_0 at the center O is the difference between fields from semicircles of radii $R_1 = 6\pi$ m and $R_2 = 4\pi$ m : $B_0 = \left|B_{R_1} - B_{R_2}\right|$.

Using $B = \dfrac{\mu_0 I}{4R}$,

we get $B_0 = \dfrac{\mu_0 I}{4}\left(\dfrac{1}{R_2} - \dfrac{1}{R_1}\right)$.

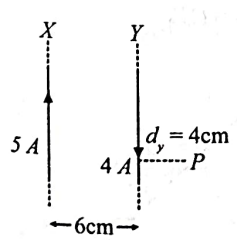
With $\mu_0 = 4\pi \times 10^{-7}$ Tm / A, $I = 12A$, $R_2 = 4\pi$, $R_1 = 6\pi$, B_0

$= \dfrac{4\pi \times 10^{-7} \times 12}{4}\left(\dfrac{1}{4\pi} - \dfrac{1}{6\pi}\right)$

$= 12\pi \times 10^{-7}\left(\dfrac{1}{12\pi}\right) = 1 \times 10^{-7}$ T.

Thus, $k = 1$.

3. [1]



$B_Y = \dfrac{\mu_0 I_Y}{2\pi d_Y}$ where d_x = 6 cm + 4 cm = 10 cm.

$= \dfrac{(4\pi \times 10^{-7}) \times 4}{2\pi \times 0.04} = 2 \times 10^{-5}$ T

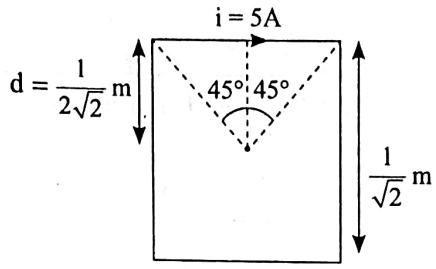
$B_X = \dfrac{\mu_0 I_X}{2\pi d_X} = \dfrac{(4\pi \times 10^{-7}) \times 5}{2\pi \times 0.1} = 1 \times 10^{-5}$ T

Using the right-hand rule, B_Y is out of the page, and B_X is into the page, so:

$B_{resultant} = B_Y - B_X = (2 - 1) \times 10^{-5}$

$= 1 \times 10^{-5}$ T

4. [8]



Square loop

The magnetic field due to single side of square loop:

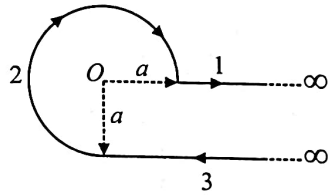
$B = \dfrac{\mu_0 i}{4\pi\, d}(\sin\theta_1 + \sin\theta_2)$

then $B = \dfrac{\mu_0 i}{4\pi\, d}(\sin\theta_1 + \sin\theta_2)$

$= \dfrac{10^{-7} \times 5 \times 2}{\frac{1}{2\sqrt{2}}} \times \dfrac{1}{\sqrt{2}} = 2 \times 10^{-6}$

$\therefore B_{net}$ at centre O = 4 B = 8×10^{-6}

5. (*b*)



$B_1 = 0$

$B_2 = \dfrac{\mu_0}{4\pi}\dfrac{i}{a}\left(\dfrac{3\pi}{2}\right)\otimes$

$B_3 = \dfrac{\mu_0 i}{4\pi a}\otimes$

$B = \dfrac{\mu_0}{4\pi}\dfrac{i}{a}\left(1 + \dfrac{3\pi}{2}\right)\otimes$

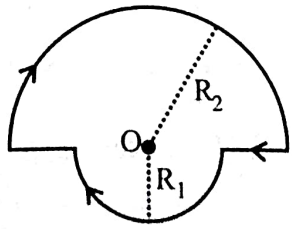
6. [160] Magnetic field point p

$B = \left(\dfrac{\mu_0 i}{2\pi a}\right) \times 2 = \dfrac{4\pi \times 10^{-7} \times 10}{\pi \times \left(\dfrac{5}{2} \times 10^{-2}\right)}$

$= 16 \times 10^{-5} = 160\mu$T

7. [3.00]

$B = \dfrac{\mu_0 i}{2R_2}\left(\dfrac{\pi}{2\pi}\right)\otimes + \dfrac{\mu_0 i}{2R_1}\left(\dfrac{\pi}{2\pi}\right)\otimes$



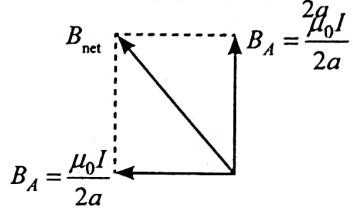
$\Rightarrow B = \left(\dfrac{\mu_0 i}{4R_2} + \dfrac{\mu_0 i}{4R_1}\right)\otimes$

$\Rightarrow B = \dfrac{4\pi \times 10^{-7} \times 4}{4 \times 4\pi} + \dfrac{4\pi \times 10^{-7} \times 4}{4 \times 2\pi}$

$= 3 \times 10^{-7} = \alpha \times 10^{-7}$

$\alpha = 3$.

8. (*c*) As we know, Magnetic field due to circular wire at its center $= \dfrac{\mu_0 I}{2a}$



Hence, $B_{net} = \dfrac{\sqrt{2}\mu_0 I}{2a}$

9. [40] Given, Current I = 5A

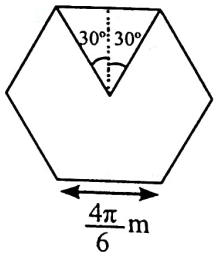
Side = 1m

$B = 4 \times \dfrac{\mu_0 i}{4\pi\left(\frac{1}{2}\right)}\left(\dfrac{1}{\sqrt{2}} + \dfrac{1}{\sqrt{2}}\right)$

$= 4 \times 10^{-7} \times 5 \times 2 \times \sqrt{2}$

$= 40\sqrt{2} \times 10^{-7}$ T

10. [72] $B = 6\left(\dfrac{\mu_0 I}{4\pi r}\right)(\sin 30° + \sin 30°)$



$= 6\dfrac{10^{-7} \times 4\pi\sqrt{3}}{\left(\dfrac{\sqrt{3} \times 4\pi}{2 \times 6}\right)} = 72 \times 10^{-7}$T

11. [5] $B_C = \dfrac{\mu_0(2i)}{2\pi r} + \dfrac{\mu_0 i}{2\pi(3r)} = \dfrac{7\mu_0 i}{6\pi r}$

$B_A = \dfrac{\mu_0 i}{2\pi r} + \dfrac{\mu_0(2i)}{2\pi(3r)} = \dfrac{5\mu_0 i}{6\pi r}$

$\therefore \dfrac{B_A}{B_C} = \dfrac{5}{7}$

$\therefore x = 5$

12. [2] Given, $\vec{r} = 0.5\hat{j}$, $\Delta l = \Delta x\hat{i}$

$\vec{dB} = \dfrac{\mu_0 I}{4\pi}\dfrac{(\vec{dl} \times \vec{r})}{r^3}$

$$\overline{dB} = \frac{10^{-7}\times 10\times\left(\frac{1}{2}\times\frac{1}{100}\right)}{\left(\frac{1}{2}\right)^3}$$

$$= 4\times 10^{-8} T$$

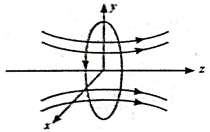
13. [2] $\vec{E}_P = \left(\frac{\sigma}{2\varepsilon_0}+\frac{2\sigma}{2\varepsilon_0}+\frac{\sigma}{2\varepsilon_0}\right)(-\hat{i})$

$$= -\frac{2\sigma}{\epsilon_0}\hat{i}$$

14. (*d*) Magnetic field at the centre of a semicircular arc is,

$$= \frac{\mu_0 I}{4R} = \frac{4\pi\times 10^{-7}\times 3}{4\times\frac{\pi}{10}} \Rightarrow B_O = 3\mu T$$

15. (*c*) Magnetic field at the center of the coil must be zero, therefore option (*a*) and (*d*) are ruled out.



Magnetic field at centre are along z direction so $B_y = 0$

For $z > 0$ B_z is + ve

For $z < 0$ B_z is − ve

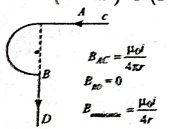
As the magnetic field to the left and to the right of the origin must be in opposite directions, so option (*c*) is correct.

16. (*d*) Magnetic field at point O due to current in BC and ET is

$$B_0 = \frac{\mu_0 I}{4\pi\frac{a}{2}} + \frac{\mu_0 I}{4\pi\frac{a}{2}} = \frac{\mu_0 I}{\pi a}$$

Due to AB & ED field is zero.

17. (*c*) $B_P = \left(\frac{\mu_0 i}{4r}+\frac{\mu_0 i}{4\pi r}\right) = \frac{\mu_0 i}{2r}\left(\frac{1}{2}+\frac{1}{2\pi}\right)$



$$B_{AC} = \frac{\mu_0 i}{4\pi r}$$

$$B_{BD} = 0$$

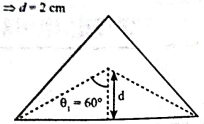
$$B_{semicircle} = \frac{\mu_0 i}{4r}$$

18. (*d*) Magnetic field due to any side of the triangle

$$B = \frac{\mu_0 I}{4\pi d}(\sin\theta_1 + \sin\theta_2)$$

$$d\tan 60^\circ = 2\sqrt{3}$$

$\Rightarrow d = 2$ cm



Total magnetic field at the centroid of the triangle

$$B = 3\times\frac{\mu_0 i}{2\pi d}\sin 60^\circ$$

$$= 3\times\frac{2\times 10^{-7}\times 2}{2\times 10^{-2}}\times\frac{\sqrt{3}}{2}, = 3\sqrt{3}\times 10^{-5} T$$

19. (*c*) Magnetic field due to a current carrying circular loop at any point on its axis is given by

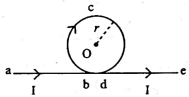
$$B = \frac{\mu_0 i r^2}{2\left(r^2+x^2\right)^{3/2}}$$

At centre, $x = 0, B_1 = \frac{\mu_0 i}{2r}$

At $x = r, B_2 = \frac{\mu_0 i}{2\times 2\sqrt{2}r}$

$$\Rightarrow \frac{B_1}{B_2} = 2\sqrt{2}$$

20. (*c*) For option (*a*)



$$B_{ab} = \frac{\mu_0}{4\pi}\frac{I}{r} \text{ (outwards)}$$

$$B_{bcd} = \frac{\mu_0}{4\pi}\frac{I}{r}(2\pi) \text{ (inwards)}$$

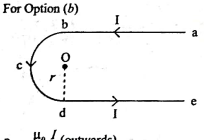
$$B_{de} = \frac{\mu_0}{4\pi}\frac{I}{r} \text{ (outwards)}$$

Hence magnetic field at O is

$$B_0 = \frac{\mu_0}{4\pi}\frac{I}{r} + \frac{\mu_0}{4\pi}\frac{I}{r}(2\pi) - \frac{\mu_0}{4\pi}\frac{I}{r}$$

$$B_0 = \frac{\mu_0}{2\pi}\frac{I}{r}(\pi - 1) \quad \text{...option (III)}$$

For Option (*b*)



$$B_{ab} = \frac{\mu_0}{4\pi}\frac{I}{r} \text{ (outwards)}$$

$$B_{bcd} = \frac{\mu_0}{4\pi}\frac{I}{r}(\pi) \text{ (outwards)}$$

$$B_{de} = \frac{\mu_0}{4\pi}\frac{I}{r} \text{ (outwards)}$$

Hence magnetic field at O is $B_0 = \frac{\mu_0 I}{4r}$ ...option (II)

21. [68]

Magnetic fields produced by both the wires will be perpendicular to each other.

$$B_1 = \frac{\mu_0 i_1}{2\pi d} \text{ and } B_2 = \frac{\mu_0 i_2}{2\pi d}$$

$$B_{net} = \sqrt{B_1^2 + B_2^2}$$

$$= \frac{\mu_0}{2\pi d}\sqrt{i_1^2 + i_2^2}$$

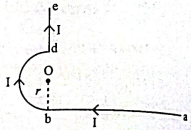
$$B_{de} = \frac{\mu_0}{4\pi}\frac{I}{r} \text{ (outwards)}$$

Hence magnetic field at O is

$$B_0 = \frac{\mu_0}{4\pi}\frac{I}{r} + \frac{\mu_0}{4\pi}\frac{I}{r}(\pi) + \frac{\mu_0}{4\pi}\frac{I}{r}$$

$$B_0 = \frac{\mu_0}{4\pi}\frac{I}{r}(\pi + 2) \text{...option (I)}$$

For option (*c*)



$$B_{ab} = \frac{\mu_0}{4\pi}\frac{I}{r} \text{ (inwards)}$$

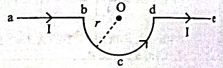
$$B_{bcd} = \frac{\mu_0}{4\pi}\frac{I}{r}(\pi) \text{ (inwards)}$$

$$B_{de} = 0 \text{ (at point } O)$$

Hence magnetic field at O is

$$B_0 = \frac{\mu_0 I}{4\pi r}(1 + \pi) \text{ ...option (IV)}$$

For option (*d*)



$$B_{ab} = 0 \text{ (at point } O)$$

$$B_{bcd} = \frac{\mu_0}{4\pi}\frac{I}{r}(\pi) \text{ (outwards)}$$

$$B_{de} = 0 \text{ (outwards)}$$

Hence magnetic field at O is $B_0 = \frac{\mu_0 I}{4r}$

22. [628]

$$B_y = |B_x| = \frac{\mu_0 I}{2r}$$

$$= \frac{4\pi\times 10^{-7}}{2\times 0.2}\times\sqrt{2}T$$

Net magnetic field at the centre,

$$B_{net} = B\sqrt{2} = \frac{4\pi\times 10^{-7}\times\sqrt{2}}{2\times 0.2}\times\sqrt{2}T$$

$$= 2\pi\times 10^{-6} N$$

$$= 628\times 10^{-6} T$$

23. [2] $B_O = B_{wire} + B_{semicircle}$

$$B_O = 0 + \frac{\mu_0 I}{4R} = \frac{4\pi\times 10^{-7}\times 14}{4\times 2.2\times 10^{-2}}$$

$$B_O = 2\times 10^{-4} T$$

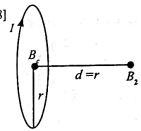
24. [40] From Biot-savart law

$$B = \frac{\mu_0}{4\pi}\frac{qv\sin\theta}{r^2} = \frac{\mu_0}{4\pi}\frac{ev}{r^2} \left(\because \sin\frac{\pi}{2} = 1\right)$$

$$B = \frac{\mu_0}{4\pi}\frac{ev\sin(\pi/2)}{r^2}$$

$$B = \frac{10^{-7}\times 1.6\times 10^{-19}\times 6.76\times 10^6}{0.52\times 0.52\times 10^{-20}} = 40T$$

25. [8]



Magnetic field due to current carrying coil at a point situated at a distance d on axial line

$$B = \frac{\mu_0 I R r^2}{2(r^2 + d^2)^{3/2}}$$

Magnetic field at cente, $d = 0$

$$B_C = \frac{\mu_0 I r^2}{2r^3} = \frac{\mu_0 I}{2r}$$

Magnetic field at $d = r$

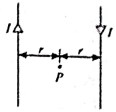
$$B_d = \frac{\mu_0 I r^2}{2(r^2 + r^2)^{3/2}} = \frac{\mu_0 I}{4\sqrt{2}\cdot r}$$

$$\frac{B_c}{B_d} = \frac{\mu_0 I}{2r}\times\frac{4\sqrt{2}r}{\mu_0 I} = \frac{2\sqrt{2}}{1} = \frac{\sqrt{8}}{1} \quad x = 8$$

26. (*b*) As we know that, if currents are flowing in same direction, magnetic field will cancel each other. Therefore, the currents must flowing in opposite direction

$$= \frac{4\pi\times 10^{-7}}{2\pi\times(7/\sqrt{2})\times 10^{-2}}\times\sqrt{8^2 + 15^2}$$

$$B_{net} = 68\times 10^{-6} T$$

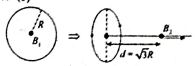


$$B_P = \frac{\mu_0 I}{2\pi r} + \frac{\mu_0 I}{2\pi r} = \frac{\mu_0 I}{2\pi r}\times 2$$

$$\Rightarrow 300\times 10^{-4} = \frac{4\pi\times 10^{-7} I}{2\pi\times 4\times 10^{-2}}\times 2$$

$$\Rightarrow I = 30 \text{ A}$$

27. (*c*)



$$B_1 = \frac{\mu_0 i}{2R}$$

$$B_2 = \frac{\mu_0 i R^2}{2\left(R^2 + (\sqrt{3}R)^2\right)^{3/2}}$$

$$= \frac{\mu_0 i R^2}{2\times 8R^3} = \frac{\mu_0 i}{8\times 2R} = \frac{B_1}{8}$$

$$\Rightarrow \text{Required ratio, } \frac{B_1}{B_2} = 8:1$$

28. (*c*) $B_C = \frac{\mu_0 I}{2r}, B_a = \frac{\mu_0 I r^2}{2\left(x^2 + r^2\right)^{3/2}}$

At $x = \frac{r}{2}$, $B_a = \frac{\mu_0 I r^2}{2\left(\frac{r^2}{4} + r^2\right)^{3/2}}$

$$= \frac{\mu_0 I r^2}{2\left(\frac{5}{4}r^2\right)^{3/2}} = \frac{\mu_0 I}{2r}\left(\frac{4}{5}\right)^{3/2}$$

$$= \frac{\mu_0 I}{2r}\left(\frac{2}{\sqrt{5}}\right)^3 = \left(\frac{2}{\sqrt{5}}\right)^3 B$$

29. (*b*) Number of turns = N

Inner radius = a, outer radius = b,

Consider an elementary portion of radius r and thickness dr.

Number of turns in elementary portion,

$$dN = \frac{N}{2\pi(b-a)}\cdot 2\pi dr = \frac{N}{(b-a)}dr$$

∴ (dB) at centre =

$$\frac{\mu_0 I dN}{2r} \Rightarrow B = \int_a^b \frac{\mu_0 I}{2r}\times\left(\frac{N}{b-a}\right)$$

$$\Rightarrow B = \frac{\mu_0 I N}{2(b-a)} ln\left(\frac{b}{a}\right)$$

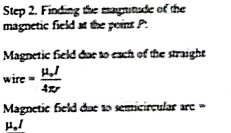
30. (*b*) B due to wire 1.

$$B_1 = \frac{\mu_0 I}{4\pi y}[\sin\theta_1 + \sin 90^\circ] \quad \text{...(i)}$$

B due to wire 2

$$B_2 = \frac{\mu_0 I}{4\pi x}[\sin\theta_2 + \sin 90^\circ]$$

So $B = B_1 + B_2$



$$B = \frac{\mu_0 I}{4\pi}\left[\frac{1}{y}\left(1 + \frac{x}{\sqrt{x^2 + y^2}}\right) + \frac{1}{x}\left(1 + \frac{y}{\sqrt{x^2 + y^2}}\right)\right] F$$

$$B = \frac{\mu_0 I}{4\pi}\left[\frac{1}{y} + \frac{1}{x} + \frac{x}{y\sqrt{x^2 + y^2}} + \frac{y}{x\sqrt{x^2 + y^2}}\right]$$

$$B = \frac{\mu_0 I}{4\pi}\left[\frac{x + y}{xy} + \frac{x^2 + y^2}{xy\sqrt{x^2 + y^2}}\right]$$

$$B = \frac{\mu_0 I}{4\pi xy}\left[(x + y) + \sqrt{x^2 + y^2}\right]$$

31. (*d*) Step 1. Given data

A hairpin is made by bending a long current-carrying wire.

Step 2. Finding the magnitude of the magnetic field at the point P:

Magnetic field due to each of the straight wire = $\frac{\mu_0 I}{4\pi r}$

Magnetic field due to semicircular arc = $\frac{\mu_0 I}{4r}$

Thus, the total magnetic field at a point P is given by

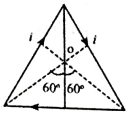
$$B = \frac{\mu_0 I}{4\pi r} + \frac{\mu_0 I}{4\pi r} + \frac{\mu_0 I}{4r}$$

$$B = \frac{\mu_0 I}{2\pi r} + \frac{\mu_0 I}{4r}$$

$$B = \frac{\mu_0 I}{4\pi r}(2 + \pi)$$

32. (*c*) $B = \frac{\mu_0 I}{4\pi r}(\sin\theta_1 + \sin\theta_2)$

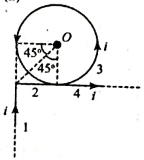
$$= \frac{10^{-7}\times 1.5}{\left(\frac{0.09}{2\sqrt{3}}\right)}\times(\sin 60^\circ + \sin 60^\circ) = 10^{-5} T$$



Net magnetic field at centroid is $B_{net} = 3 \times 10^{-5}$ T

inside the plane

33. (*d*)



The net magnetic field

$$\frac{\mu_0 i}{4\pi R}\left[\sin 90^\circ - \sin 45^\circ\right]\otimes + \frac{\mu_0 i}{2R}\odot$$

$$+\frac{\mu_0 i}{4\pi R}\left(\sin 45^\circ + \sin 90^\circ\right)\odot$$

$$\frac{-\mu_0 i}{4\pi R}\left[1-\frac{1}{\sqrt{2}}\right]+\frac{\mu_0 i}{2R}$$

$$+\frac{\mu_0 i}{4\pi R}\left[\frac{1}{\sqrt{2}}+1\right]\odot$$

$$\frac{\mu_0 i}{4\pi R}\left[\sqrt{2}+2\pi\right]\odot = \frac{\mu_0 i}{2\pi R}\left[\frac{1}{\sqrt{2}}+\pi\right]\odot$$

34. (*b*) $B_A = \frac{(\mu_0)(2)}{(2)(2)}\frac{3(\pi/2)}{2\pi}$ ...(*i*)

$B_B = \frac{(\mu_0)(3)}{(2)(4)}\frac{5\pi/3}{2\pi}$ ...(*ii*)

From (i) and (ii) $\frac{B_A}{B_B} = \frac{6}{5}$

35. (*d*) $B_C = \frac{\mu_0 i}{2R}$, $B_C = \frac{\mu_0 Ni}{2(R/N)}$

$\therefore \frac{B_L}{B_C} = \frac{1}{N^2}$

36. (*c*) $B = 2\left[\frac{\mu_0 i}{4\pi d}(\cos\theta_1 - \cos\theta_2)\right]$

$B = 10^{-4}$

$\mu_0 = 4\pi \times 10^{-7}$, $\theta_1 = 90°$, $\theta_2 = 180°$

$d = 4 \times 10^{-2}$

$\Rightarrow = 20$ A (into the page)

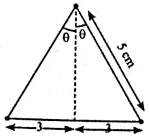
37. (*d*) $B = \frac{\mu_0 i}{4\pi d}2\sin\theta$

$d = 4$ cm

$\sin\theta = \frac{3}{5}$

$B = 10^{-7} \times \frac{5\times 2}{4\times 10^{-2}}\left(\frac{3}{5}\right)$

$B = 1.5 \times 10^{-5}$ T

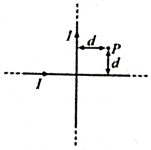


38. (*d*) $B = \frac{\mu_0 i}{2R} = \frac{\mu_0 q\omega}{2R2\pi} \Rightarrow q = \frac{B.4\pi R}{\mu_0\omega}$

$$q = \frac{(3.8\times 10^{-4})(4\pi\times 10^{-1})}{(4\pi\times 10^{-7})(40\pi)}$$

$q = 3 \times 10^{-3}$ C

39. (*a*)



Magnetic field at point P

$$\vec{B}_{net} = \frac{\mu_0 I}{2\pi d}(-\hat{k}) + \frac{\mu_0 I}{2\pi d}(\hat{k}) = 0$$

40. (*d*) Magnetic field at centre due to current carrying loop subtending angle θ at the centre, $B = \frac{\mu_0 I}{4\pi r}\theta$.

Hence, θ is in radian.

$$B = \left(\frac{\mu_0}{4\pi}\right)\times 10\left(\frac{1}{3\times 10^{-2}} - \frac{1}{5\times 10^{-2}}\right)\frac{\pi}{4}$$

$$\Rightarrow B = \frac{\pi}{40}\times 10^{-4} = \frac{\pi}{4}\times 10^{-5} \approx 10^{-5}$$

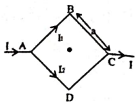
41. (*c*) The magnetic field at the center of a circular coil of radius *R* carrying current *I* is $B_1 = \frac{\mu_0 I}{2R}$. At an axial distance $x = 3R$, the field is $B_2 = B_1 \sin\theta$, where $\sin\theta = \frac{R}{\sqrt{R^2+(3R)^2}} = \frac{R}{2R} = \frac{1}{2}$. Thus, $B_2 = \frac{\mu_0 I R^2}{2(10R^2)}$, and the ratio $\frac{B_2}{B_1} = \frac{1R}{2}\frac{4R}{125}$

42. (*c*)

The loop is divided into two branches:

Branch ABC (resistance = *r*)

Branch ADC (resistance = 2*r*)



Let the total current be *I*. Since the branches are in parallel:

$I_1(r) = I_2(2r) \Rightarrow I_1 = 2I_2$

Also,

$I_1 + I_2 = I \Rightarrow 2I_2 + I_2 = I \Rightarrow I_2 = \frac{I}{3}$

and $I_1 = \frac{2I}{3}$.

The magnetic field at the center of a finite wire segment of length *a* carrying current *i* is: $B = \frac{\mu_0 i}{4\pi d}(\sin\theta_1 + \sin\theta_2)$

For the square loop:

Distance from center to any side $(d) = \frac{a}{2}$

$\theta_1 = \theta_2 = 45°$, so $\sin\theta_1 + \sin\theta_2 = \sqrt{2}$.

AB (current $I_1 = \frac{2I}{3}$):

$$B_{AB} = \frac{\mu_0(2I/3)}{4\pi(a/2)}\sqrt{2} = \frac{\sqrt{2}\mu_0 I}{3\pi a}$$

Direction: Into the plane $(-\hat{k})$.

BC (current $I_1 = \frac{2I}{3}$):
Same as AB:

$$B_{BC} = \frac{\sqrt{2}\mu_0 I}{3\pi a}(-\hat{k})$$

CD (current $I_2 = \frac{I}{3}$):

$$B_{CD} = \frac{\mu_0(I/3)}{4\pi(a/2)}\sqrt{2} = \frac{\sqrt{2}\mu_0 I}{6\pi a}$$

Direction: Out of the plane $(+\hat{k})$.

DA (current $I_2 = \frac{I}{3}$):

Same as CD:

$$B_{DA} = \frac{\sqrt{2}\mu_0 I}{6\pi a}(+\hat{k})$$

$$B_{resultant} = -B_{AB} - B_{BC} + B_{CD} + B_{DA}$$

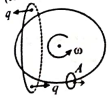
$$= -\frac{\sqrt{2}\mu_0 I}{3\pi a} - \frac{\sqrt{2}\mu_0 I}{3\pi a} + \frac{\sqrt{2}\mu_0 I}{6\pi a} + \frac{\sqrt{2}\mu_0 I}{6\pi a}$$

$$= -\frac{2\sqrt{2}\mu_0 I}{3\pi a} + \frac{\sqrt{2}\mu_0 I}{3\pi a} = -\frac{\sqrt{2}\mu_0 I}{3\pi a}$$

$$|B_{resultant}| = \frac{\sqrt{2}\mu_0 I}{3\pi a}$$

43. (*c*) $I_A = \frac{\text{Total charge}}{\text{Time}} = \frac{Nq}{\frac{2\pi}{\omega}} = \frac{Nq\omega}{2\pi}$

(for Amperian loop 'A')



But, $I_B = 0$ (for Amperian loop 'B')

So, $I_A - I_B = \frac{Nq\omega}{2\pi}$

44. (*a*) For $r < R$, $B = \frac{\mu_0 Ir}{2\pi R^2}$

$B \propto r$

and for $r > R$, $B = \frac{\mu_0 I}{2\pi r}$

$B \propto \frac{1}{r}$

45. [250] Time period is given by $\frac{2\pi m}{qB}$

Magnetic field *B* in solenoid = $\mu_0 nI$

$T = \frac{2\pi m}{q(\mu_0 nI)}$

T = 75 *ns*

$$75\times 10^{-9} = \frac{(2\pi)(9\times 10^{-31})}{1.6\times 10^{-19}\times 4\pi\times 10^{-7}\times n\times 1.5}$$

$\Rightarrow n = 250$

46. (*d*) $\frac{B_{max}}{2} = \frac{\mu_0 I}{4\pi a}$

It can be obtained inside as well as outside the wire

For inside,

$\frac{\mu_0 I}{4\pi a} = \frac{\mu_0 Ir}{2\pi a^2} \Rightarrow r = \frac{a}{2}$

For outside

$\frac{\mu_0 I}{4\pi a} = \frac{\mu_0 I}{2\pi r} \Rightarrow r = 2a$

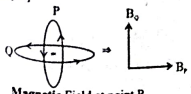
47. [8] Magnetic field inside solenoid

$B = \mu_0 ni \left(n = \frac{N}{l}\right)$

$\Rightarrow i = \frac{\mu_0 Ni}{B} = \frac{(4\pi\times 10^{-7})(200)(0.29)}{2.9\times 10^{-4}}m$

$= 8\pi$ cm

48. [20]



Magnetic Field at point P,

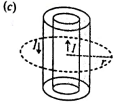
$B_P = \frac{\mu_0 Ni_1}{2r} = \frac{\mu_0\times 1\times 100}{2\pi} = 2\times 10^{-3}T$

$B_Q = \frac{\mu_0 Ni_2}{2r} = \frac{\mu_0\times 2\times 100}{2\pi} = 4\times 10^{-3}T$

$B_{net} = \sqrt{B_P^2 + B_Q^2} = \sqrt{20}$mT

$x = 20$

49. (*c*)



Consider loop as shown

here $\oint \vec{B}.d\vec{l} = \mu_0 I_{enc} = 0$

$\therefore B = 0$ outside the cable

50. [500]

Magnetic field the inside the solenoid, $B = \mu_0 ni$

$\mu_0\left(\frac{m}{\ell}\right)i = B$

$m = \frac{B\ell}{\mu_0 i} = \frac{6.28\times 10^{-3}\times 0.5}{12.56\times 10^{-7}\times 5}$

m = 500

51. (*c*) From circuital law

$B_1 2\pi\frac{a}{2} = \mu_0\frac{I}{4}$

$B_1 = \frac{\mu_0 I}{4\pi a}$ (I)

$B_2 2\pi 2a = \mu_0 I$

$B_2 = \frac{\mu_0 I}{4\pi a}$ (II)

From (i) and (ii) we get the required ratio 1 : 1

52. (*b*) Statement I is true but Statement II is false

53. (*b*) $H = \frac{B}{\mu_0} = \frac{\mu_0 ni}{\mu_0} = ni \Rightarrow i = \frac{H}{n}$

$i = \frac{1.6\times 10^3}{\left(\frac{8}{10^{-2}}\right)} = 0.2\times 10^3\times 10^{-2} = 2A$

54. (*c*) It is given that, Final kinetic energy $(K.E_f)$ = 4 times Initial kinetic energy $(K.E_i)$

We know that $r \propto \sqrt{k\cdot E}$

$\therefore \frac{r_f}{r_0} = \frac{\sqrt{k\cdot E_f}}{k\cdot E_i}$

$\frac{r_f}{r_0} = \frac{\sqrt{4k.E_i}}{\sqrt{k.E_i}}$

$\frac{r_f}{r_0} = \frac{2}{1} = 2:1$

55. (*a*) We know that, $B = n\mu_0 I$

According to equations $I' = 2I$

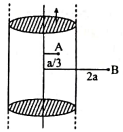
$n' = n/2$

$B' = \frac{n}{2}\mu_0 2I = B$

56. (*d*) $B_A = \frac{\mu_0 ir}{2\pi a^2} = \frac{\mu_0 i}{2\pi a^2}\frac{a}{3}$

$B_B = \frac{\mu_0 i}{2\pi(2a)}$

$\frac{B_A}{B_B} = \frac{4}{6} = \frac{2}{3}$



57. (*d*)

58. (*d*) A magnetic field does not work on the speed of a charged particle, only altering its direction via the Lorentz force. Since speed remains constant, the de-Broglie wavelength, which depends on speed $\left(\lambda = \frac{h}{mv}\right)$, stays unchanged. Thus, it remains λ_0.

59. (*a*) Radius (R) = Mv/QB = P/QB

$R \propto \frac{1}{Q}$

Assertion is true reason is false

In above picture replace 'r' with 'R' and 'q' with 'Q'

60. [0] Motion will be uniform circular motion, therefore, time period of circular path:

$T = \frac{2\pi m}{qB}$

where:

m = 16 μg = 16 × 10⁻⁹ kg (mass of the particle)

q = 1.6 μC = 1.6 × 10⁻⁶ C (charge of the particle)

B = 6.28 T (magnetic field strength)

So,

$$T = \frac{2\times 3.14\times 16\times 10^{-9}}{1.6\times 10^{-6}\times 6.28}$$

The particle completes one full revolution in 0.01 seconds.

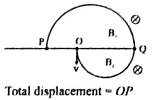
The nearest integer to 0.01 is 0.

NTA Answer is 10

Correct Answer is 0 (nearest integer)

61. (*d*) Particle starts from O moves in semicircle to come to Q & then again covers semicircle to reach P.

As $\vec{v}$ is $\perp$ to $\vec{B}$, so charge particle will move in circular path, whose radius is given by $R = \frac{mv}{qB}$



Total displacement = OP

$PQ - OQ$

$$PO = \frac{2mv}{qB_1} - \frac{2mv}{qB_2}$$

$$PO = \frac{2mv}{qB_1}\left[1 - \frac{B_1}{B_2}\right]$$

62. (*d*) radius is given by $r = \frac{mv}{qB}$

$$\Rightarrow v = \frac{qBr}{m}$$

kinetic energy

$$E = \frac{1}{2}mv^2 \propto v^2$$

$$E \propto \frac{q^2B^2r^2}{m^2}$$

$\alpha\ r^2$

$E\ \alpha\ r^2$

63. [2] Using the centripetal force equation:

$qvB = \frac{mv^2}{r}$ (for circular path)

Solving for B: $B = \frac{mv}{qr}$

$$B = \frac{(1.6\times 10^{-27})(2\times 10^5)}{(1.6\times 10^{-19})(2\times 10^{-2})}$$

$$B = \frac{3.2\times 10^{-22}}{3.2\times 10^{-21}} = 0.1\ T$$

For no deflection, the electric force equals the magnetic force:

$qE = qvB$ (for No defection

$E = vB$

$E = (2\times 10^5)(0.1)$

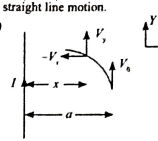
$E = 2\times 10^4$ N/C

64. (*b*) (A) is true

Magnetic force

$\vec{F} = q(\vec{v}\times\vec{B}) = 0$ if $\vec{v}$ is parallel to $\vec{B}$ so straight line motion.

65. (*d*)



$\vec{V} = V_x\hat{i} + V_y\hat{j}$

$\vec{F} = q\vec{V}\times\vec{B}$

$= q(-V_x\hat{i} + V_y\hat{j})\times B(-\hat{k})$

$\vec{F} = -qBV_x\hat{j} + qBV_y(-\hat{i})$

So, $F_x = -qBV_x$

$m\frac{dv_x}{dt} = -qBV_x$

$m\frac{dV_y}{dx}\left(\frac{dx}{dt}\right) = -qB\left(-\frac{dx}{dt}\right)$

$dV_y = \frac{qB}{m}dx$

$dV_y = \frac{q\mu_0 I}{2\pi xm}dx$

$\int_{V_0}^{-V_0} qV_y = \frac{q\mu_0 I}{2\pi m}\int_a^{x_0}\frac{dx}{x}$

$-2V_0 = \frac{q\mu_0 I}{2\pi m}\ln\left(\frac{x_0}{a}\right)$

$\ln\left(\frac{x_0}{a}\right) = -\frac{4\pi mV_0}{\mu_0 qI}$

$x_0 = ae^{-\frac{4\pi mV_0}{\mu_0 qI}}$

66. (*c*) $\vec{F}_E = q\vec{E}$ and $\vec{F}_B = q(\vec{V}\times\vec{B})$

Net force on particle must be zero i.e.

$q\vec{E} + q\vec{V}\times\vec{B} = 0$

Possible cases are

(i) $\vec{E}\ \&\ \vec{B} = 0$

(ii) $\vec{V}\times\vec{B} = 0, \vec{E} = 0$

(iii) $q\vec{E} = -q\vec{V}\times\vec{B}$

$\vec{E} \neq 0\ \&\ \vec{B} \neq 0$

67. (*b*) Radius of circular path

$$R = \frac{mv}{qB} = \frac{p}{qB} = \frac{\sqrt{2m(KE)}}{qB} = \frac{\sqrt{2mqV}}{qB}$$

$R \propto \sqrt{m}$

$m \propto R^2$

mass ratio of X and Y is

$$\frac{m_1}{m_2} = \left(\frac{R_1}{R_2}\right)$$

68. [32] $q = 4\ \mu C$, $\vec{v} = 4\times 10^6\hat{j}$ m/s

$\vec{B} = 2\hat{k}T$

The force acting on the charge is

$F = q(\vec{v}\times B) = 4\times 10^{-6}(4\times 10^6\hat{j}\times 2\hat{k})$

$= 4\times 10^{-6}\times 8\times 10^6\hat{i}$

$\vec{F} = 32\hat{i}$ N

$x = 32$

69. [5] Lorentz Force, $\vec{F} = q(\vec{v}\times\vec{B})$

$5e\hat{k} = e(3\hat{i} + 5\hat{j})\times(B_0\hat{i} + 2B_0\hat{j})$

$5e\hat{k} = e(6B_0\hat{k} - 5B_0\hat{k})$

$\Rightarrow B_0 = 5T$

70. (*b*) Since $\vec{v}$ is parallel to $\vec{B}$ hence force on electron due to magnetic field is zero. So electron will move along axis with uniform velocity.

71. (*c*) The electrostatics force on a charge particle $= q\vec{E}$

The magnetic force on the charge particle $= q(\vec{V}\times\vec{B})$

72. (*d*) Net force on the electron should be zero so that it moves along the same path.

$qE = qVB$

$E = VB$

$= \sqrt{\frac{2\times KE}{m}}\times B$

$= \sqrt{\frac{2\times 5\times 1.6\times 10^{-19}}{9\times 10^{-31}}}\times 3\times 10^{-6} = 4N/C$

73. (*c*) Hence magnetic field is uniform, now radius of the path,

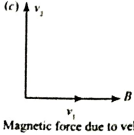
$$R = \frac{mv}{qB} = \frac{\sqrt{2m(K.E)}}{qB} \Rightarrow R \propto \frac{\sqrt{m}}{q}$$

$$\therefore \frac{R_{deuteron}}{R_{proton}} = \sqrt{\frac{m_d}{m_p}}\times\frac{q_p}{q_d} = \sqrt{2}\times 1$$

The ratio of the radius of deutron and proton path,

$\therefore r_d : r_p = \sqrt{2} : 1$

74. (*c*)



Magnetic force due to velocity component v_1

$F = qv_1 B \sin 0 = 0$

So v_1 will same.

Because of velocity component v_2, magnetic force act towards centre i.e. moving it circular.

Hence path is helical with the axis along magnetic field B.

75. (*a*) Force on the $\vec{e}$, $\vec{F} = -e(\vec{v}\times\vec{B})$

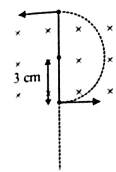
Form right hand thumb rule force will be along -ve y-axis.

Since, magnetic force is $\perp$ to velocity, therefore path of electron would be a circle.

76. (*b*) $\vec{F} = q(\vec{v}\times\vec{B})$ as angle between $\vec{v}$ and $\vec{B}$ is $0°$ $\vec{F} = 0$. $\vec{e}$ will not experience force. In the solenoid, magnetic field lie along the axis of solenoid.

77. [144] $r = \frac{mv}{qB} = \frac{\sqrt{2km}}{qB}$, $m = \frac{r^2q^2B^2}{2k}$

$m = \frac{r^2q^2B^2}{2k}$



$$\frac{1}{100}\times\frac{3}{100}\times 2\times 2\times 4$$

$$m = \frac{\times 10^{-3}\times 4\times 10^{-3}\times 10^{-12}}{2\times(100)\times 10^{-6}}$$

$m = 144\times 10^{-18}$ kg

78. [40]

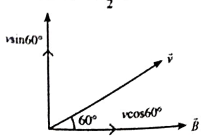
$B = \frac{\pi}{2}\times 10^{-3}$

$v = \sqrt{\frac{2KE}{m}} = \sqrt{\frac{2\times 2\times 1.6\times 10^{-19}}{1.6\times 10^{-27}}}$

$v = 2\times 10^4$ m/sec

Time period of one rotation $= \frac{2\pi m}{eB}$

$$T = \frac{2\pi\times 1.6\times 10^{-27}}{1.6\times 10^{-19}\times\frac{\pi}{2}\times 10^{-3}} = 4\times 10^{-5}\ sec$$



Pitch = $(v\cos 60°)T$

Pitch $= 2\times 10^4\times\frac{1}{2}\times 4\times 10^{-5} = 40$ cm

79. (*a*) The magnetic force acts perpendicularly to the velocity of the particle. Thus, no work is done by the force on the particle. Hence, speed and energy of the particle remains constant.

80. (*b*) $\vec{a}\cdot\vec{B} = 0$

$(\alpha\hat{i} - 4\hat{j})\cdot(2\hat{i} + 3\hat{j}) = 0$

$\alpha(2) + (-4)(3) = 0$

$\alpha = \frac{12}{2} \Rightarrow \alpha = 6$

81. (*d*) We know that, $r = \sqrt{\frac{2mKE}{qB}}$

$\therefore r \propto \sqrt{\frac{m}{q}}$

Let us suppose,

$m_p = m, q_p = q, m_d = 2m, q_d = q, m\alpha = 4m, q\alpha = 2q.$

$\therefore r_p : r_d : r_\alpha = \sqrt{\frac{m_p}{q_p}} : \sqrt{\frac{m_d}{q_d}} : \sqrt{\frac{m_\alpha}{q_\alpha}}$

$= \sqrt{\frac{m}{q}} : \sqrt{\frac{2m}{q}} : \sqrt{\frac{4m}{2q}} = 1 : \sqrt{2} : 1$

$\therefore r_p : r_d : r_\alpha = 1 : \sqrt{2} : 1$

82. (*b*) Radius of circular path $R = \frac{\sqrt{2mE}}{qB}$

$q = \frac{\sqrt{2mE}}{RB}$

Kinetic energy and magnetic field are constant, therefore, $q \propto \frac{\sqrt{m}}{R}$

$$\Rightarrow \frac{q_1}{q_2} = \sqrt{\frac{m_1}{m_2}}\times\frac{R_2}{R_1} = \sqrt{\frac{9}{4}}\times\frac{5}{6} = \frac{5}{4}$$

83. (*c*)

$$f = \frac{1}{T} = \frac{qB}{2\pi m} = \frac{1.6\times 10^{-19}\times 1.0\times 10^{-4}}{2\times 3.14\times 9\times 10^{-31}}$$

$= 0.025\times 10^4$ Hz $= 2.8\times 10^6$ Hz

84. (*c*) $\frac{r_\alpha}{r_p} = \frac{m_\alpha q_p}{m_p q_\alpha}$

($\because r = \frac{mv}{qB}$, v & B are constant)

$$= \frac{4m}{m}\left(\frac{q}{2q}\right) = 2:1$$

85. (*c*) When a particle moves in a uniform magnetic field, The force due to magnetic field on it is perpendicular to its velocity, Because, Force $\vec{F} = q(\vec{v}\times\vec{B})$

Since the magnetic field is constant, so force is constant. Hence, its direction of velocity constantly changes but magnitude of velocity remains same.

Hence the particle moves in circular motion Since magnitude of velocity does not changes, Hence kinetic energy also does not change.

Statement-II is incorrect because the velocity of the particle is affected by magnetic field. Its direction changes

86. (*c*) $E_y = 900\sin\omega\left(t - \frac{x}{c}\right)$

Comparing with $E_y = E_0\sin\omega\left(t - \frac{x}{c}\right)$, $E_0 = 900$

Given, speed of charge, $v = 3\times 10^7$ m/s

$$\therefore \frac{F_E}{F_B} = \frac{qE}{|vB_0|} = \frac{c}{V} = \frac{3\times 10^8}{3\times 10^7} = 10$$

$$\Rightarrow \frac{F_E}{F_B} = \frac{10}{1}$$

87. (*a*) $KE = 728\times 1.6\times 10^{-19} = \frac{1}{2}\times 9.1\times 10^{-31}\times v^2$

$\Rightarrow v = 16\times 10^6$ m/s

For lorentz force to be zero

$\Rightarrow eE = eVB$

$\Rightarrow E = vB = 16\times 10^6\times 12\times 10^{-3}$

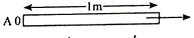
$= 192\times 10^3$ V/m

88. [136] Given, current, $i = 1A$

Area, $A = 2mm^2$

$\rho = 1.7\times 10^{-8}\ \Omega m$

Resistivity, $= 1.7\times 10^{-8}\ \Omega m$



$\because J = \sigma E$, $\sigma = \frac{1}{\rho}$, and $J = \frac{I}{A}$

Therefore, Force on e^-,

$$F = eE = \frac{ej}{\sigma} = \frac{ei}{A\sigma} = \frac{ei\rho}{A}$$

$$= \frac{1.6\times 10^{-19}\times 1.7\times 10^{-8}}{2\times 10^{-6}}$$

$= 0.8\times 1.7\times 10^{-21}$

$= 136\times 10^{-23}$ N

89. [2] Since, radii of circular path,

$r = \frac{mv}{qB}$

$\because p = mv = \sqrt{2m(K.E)}$

$\therefore r = \frac{\sqrt{2m(K.E)}}{qB}$

For constant K.E, $r \propto \frac{\sqrt{m}}{q}$

$$\frac{r_d}{r_p} = \frac{\sqrt{m_d}}{\sqrt{m_p}}\times\frac{q_p}{q_d}$$

$$= \frac{\sqrt{2m_p}}{\sqrt{m_p}}\times\frac{q_p}{q_d} \quad [q_p = q_d = 1.6\times 10^{-19}C]$$

$= \sqrt{2} : 1$

90. [10]

$$r = \frac{mv}{qB} = \frac{\sqrt{2m \cdot K}}{qB}$$

$$= \frac{\sqrt{2\times 24\times 1.66\times 10^{-27}\times 5\times 10^3\times 1.6\times 10^{-19}}}{1.6\times 10^{-19}\times 0.5}$$

$= 9.975 \times 10^{-2}$ cm

$r \approx 10$ cm

91. (*a*) We know that, charge on deutron and alpha particle is *e* and 2*e* respectively. Mass of deutron and alpha particle is 2m and 4m respectively.

As we know that radius of circular path in magnetic field is given as

$$r = \frac{mv}{qB} = \frac{\sqrt{2mK}}{qB}$$

$$\frac{r_d}{r_\alpha} = \sqrt{\frac{m_d}{m_\alpha}}\times\frac{q_\alpha}{q_d} = \frac{1}{\sqrt{2}}\times 2 = \sqrt{2}$$

92. (*d*) As we know that radius of trajectory is given as

$$R = \frac{\sqrt{2mK}}{qB} \Rightarrow K = \frac{R^2q^2B^2}{2m}$$

$$\frac{K_p}{K_\alpha} = \frac{R_p^2q_p^2B^2}{2m_p}\times\frac{2m_\alpha}{R_\alpha^2q_\alpha^2B^2}$$

$$\Rightarrow \quad = \left(\frac{m_\alpha}{m_p}\right)\times\left(\frac{R_p}{R_\alpha}\right)^2\times\left(\frac{q_p}{q_\alpha}\right)^2$$

$$\Rightarrow \frac{K_p}{K_\alpha} = (4)\times(2)^2\times\left(\frac{1}{2}\right)^2 = 4:1$$

93. (*c*) Radius of a charge particle moving in circular path in perpendicular magnetic field can be given as,

$$R = \frac{\sqrt{2Em}}{B.q}$$

$$\Rightarrow R \propto \frac{\sqrt{m}}{q}$$

$$\Rightarrow \frac{R_1}{R_2} = \frac{\sqrt{4}}{\sqrt{16}}\times\frac{3}{2} = \frac{3}{4}$$

$$\Rightarrow R_2 = \frac{4R_1}{3}$$

$$\sin\theta = \frac{d}{R} \Rightarrow \theta\alpha\frac{1}{R} \Rightarrow \theta_2 < \theta_1$$

94. (*b*) Magnetic force acting on a charge particle is

$$\vec{F} = q(\vec{v}\times\vec{B})$$

$$W = \vec{F}\cdot d\vec{r}$$

Since, magnetic force acting on point charge and displacement will be always perpendicular.

So, $W = F.S$

$\Rightarrow W = 0$

95. (*c*) Given, $\frac{q_1}{q_2} = \frac{1}{2}, \frac{v_1}{v_2} = \frac{2}{3}$

Radius of the circular trajectory in the magnetic field.

$$r = \frac{mv}{qB}, \therefore \frac{r_1}{r_2} = \left(\frac{m_1}{m_2}\right)\left(\frac{q_2}{q_1}\right)\left(\frac{v_1}{v_2}\right)$$

$$\frac{r_1}{r_2} = 2\times\frac{2}{3} = \frac{4}{3}$$

96. (*a*) Charge

Mass

Proton e m

α-particle 2e 4m

Deuteron e 2m

Speed, $v = P/m$

$$v_1 = \frac{P}{m}, v_2 = \frac{P}{2m}, v_3 = \frac{P}{4m}$$

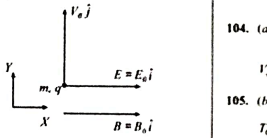
$$\therefore v_1 : v_2 : v_3 = 4:2:1$$

$$F = qvB = \frac{qPB}{m}$$

$$F_1 = \frac{qPB}{m}, F_2 = \frac{q\times P\times B}{2m}, F_3 = \frac{2qPB}{4m}$$

$$\therefore F_1 : F_2 : F_3 = 4:2:2 = 2:1:1$$

97. (*d*)



$$(2V_e)^2 = v_e^2 + v_e^2; v_e = \sqrt{3}v_e$$

$$\therefore \sqrt{3}v_e = 0 + \frac{eE}{m}t \Rightarrow t = \frac{mv_e\sqrt{3}}{qE}$$

98. (*c*) Force an the electron,

$$F = q(\vec{E}+\vec{V}\times\vec{B}) \cdot \vec{E}+\vec{V}\times\vec{B} = 0$$

99. (*a*) $t = 10 \cdot$ pitch

$$= 10\times v\cos 60°\times\frac{2\pi m}{qB}$$

$$= \frac{20\times 3.14\times 1.67\times 10^{-27}\times 4\times 10^6\times\frac{1}{2}}{1.6\times 10^{-19}\times 0.3}$$

$= 4.36 \cdot 10^{-1}$ m ≈ 0.44 m.

100. (*d*) $F = q(\vec{V}\times\vec{B})$

$= (1 \times 10^{-4})$

$[2\hat{i}+3\hat{j}+4\hat{k}]\times[5\hat{i}+3\hat{j}-6\hat{k}]\times 10^{-3}$

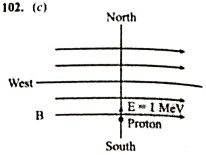
$= (-30\hat{i}+32\hat{j}-9\hat{k})10^{-4}$ N

101. (*c*) The cyclotron equation is $r = \frac{mv}{qB_0}$

When charged particle should not hit the screen

$$\Rightarrow \frac{mv}{qB_0} < d, \therefore V_{max} = \frac{qB_0d}{m}$$

102. (*c*)



$$\because K.E. = 1.6\times 10^{-13} = \frac{1}{2}\times 1.6\times 10^{-27}v^2$$

$v = \sqrt{2}\times 10^7$ m/s

$\therefore Bqv = ma$

$$B = \frac{1.6\times 10^{-27}\times 10^{12}}{1.6\times 10^{-19}\times\sqrt{2}\times 10^7}$$

$= 0.71 \times 10^{-3}$ T

So, 0.71 mT

103. (*d*) Pitch $= v\cos\theta \times \frac{2\pi m}{qB}$

$$= 4\times 10^7\times\frac{1}{2}\times\frac{2\pi\times 1.67\times 10^{-27}}{1.69\times 10^{-19}\times 0.3} = 4 \text{ cm}$$

104. (*a*) $R_{max} = \frac{R}{2} = \frac{mv_{max}}{eB_0nl}$

$$V_{max} = \frac{Re\mu_0nl}{2m}$$

105. (*b*) Time period of oscillation,

$$T_0 = 2\pi\sqrt{\frac{m}{k}} = \frac{2\pi}{\sqrt{10}}$$

For damped oscillation, amplitude is given as

$A = A_0e^{-\gamma t}$

for $A = \frac{A_0}{e}, t = \gamma$

$$t = \gamma = \frac{2m}{b} = \frac{2m}{B^2l^2} = 10^4 \text{ s}$$

$\therefore$ No of oscillation,

$$N = \frac{t}{T_0} = \frac{10^4}{2\pi/\sqrt{10}} = 5000$$

106. (*a*) $\vec{t} = \vec{M}\times\vec{B}$

$\Rightarrow |\vec{t}| = MB\sin 90° = C\theta$

$\Rightarrow MB = C\theta \Rightarrow C\theta = i\,N\,A\,B$

$$\Rightarrow 10^{-4}\times\frac{\pi}{180} = 10^{-3}\times 10^{-4}\times 175\times B$$

$\Rightarrow B = 10^{-3}$Tesla.

107. (*c*) $r = \frac{mv}{qB} = \frac{\sqrt{2mK}}{qB}$

$\Rightarrow r_{He} = r_p > r_e$

108. (*) $\hat{a} = -\left(\frac{d\sqrt{3}\hat{i}+d\hat{j}}{\sqrt{(d\sqrt{3})^2+d^2}}\right) = \frac{-\sqrt{3}\hat{i}-\hat{j}}{2}$

$\therefore$ Acceleration,

$$\vec{a} = \frac{F}{m}\hat{a} = \frac{qVB}{m}\left(\frac{-\sqrt{3}\hat{i}-\hat{j}}{2}\right) \text{ (Bonus)}$$

109. (*a*) $m_r = m$

$q_r = q, K_r = qV = K, m\alpha = 4m, q_\alpha = 2q$

$K_\alpha = 2qV = 2K$

Radius of circular path,

$$r = \frac{mv}{qB} = \frac{\sqrt{2Km}}{qB} = \sqrt{\frac{2mV}{qB^2}} \Rightarrow \frac{r_p}{r_\alpha} = \frac{1}{\sqrt{2}}$$

110. (*d*) Radius, $r = \frac{mv}{qB} = \frac{\sqrt{2mqv}}{qB}$

$$r = \frac{\sqrt{\frac{2mv}{q}}}{B} = \frac{\sqrt{\frac{2\times 9.1\times 10^{-31}\times 500}{1.6\times 10^{-19}}}}{100\times 10^{-3}}$$

$r = 7.5 \times 10^{-4}$ m

111. [48] The force F on a wire carrying current I in a magnetic field B is $F = ILB$, with L as wire length. For $I = 8$ A, $L = 4.0$ cm $= 0.04$ m,

and B = 0.15T, we get $F = 8 \times 0.04 \times 0.15 = 0.048$ N = 48mN.

112. [35] Given, $B_H = 3.5\times 10^{-4}$ T and $\theta = 90°$

As we know, $F = iLB \sin\theta$, $1 = \sqrt{2}A$

The force per unit length is,

$$\frac{F}{l} = iB\sin\theta = \sqrt{2}\times 3.5\times 10^{-5}\times\frac{1}{\sqrt{2}}$$

113. (*d*)

$\vec{F} = i\vec{l}\times\vec{B}$ (Direction of magnetic field is in $\hat{k}$)

$l = 2R$

$\vec{F} = -2iRB\hat{j}$

114. [50] Here F_{net} will be due to portion parallel to *y*-axis, because force on segment parallel to *x*-axis will cancel each other.

$F = 0.5 \times 0.5 \times 6 \times 0.2 - 0.5 \times 0.5 \times 0.2 \times 5$

$= 0.5 \times 0.5 \times 0.2 = 50 \times 10^{-3}$ N

$= 50$ mN

115. [5] Resistance of the current carrying wire,

$$R = \frac{\rho l}{A} \Rightarrow \frac{2\times 10^{-6}\times l}{10^{-5}} = 1$$

$\Rightarrow l = 5$m

Net force in the vertical direction is zero, so

$mg = Bil$

$$B = \frac{mg}{il} = \frac{5}{2\times 5} = 0.5$$

$= 5 \times 10^{-1}$ Tesla

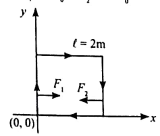
116. [160]

Given; $l = 2m$ and $i = 2A$

$B(x = 0) = B_0, B(x = 2) = 9B_0$

Also, $F = ilB$

$\Rightarrow F_1 = ilB_0$ & $F_2 = 9ilB_0$



$F = F_2 - F_1 = 8ilB_0 = 8 \times 2 \times 2 \times 5$

$F = 160$ N

117. (*a*) Force per unit length between two parallel straight current carrying wires $= \frac{\mu_0 i_1 i_2}{2\pi d}$

$$\Rightarrow \frac{F_1}{F_2} = \frac{\frac{2\pi(5cm)}{\mu_0(10)^2}}{\frac{\mu_0(20)^2}{2\pi\left(\frac{5cm}{2}\right)}} = \frac{1}{8}$$

$\Rightarrow F_2 = 8F_1$

118. (*d*) $F_m = mg$

$\therefore ILB = mg$

$$\therefore \left(\frac{V}{R}\right)LB = mg, \therefore V = \frac{mgR}{LB}$$

$$= \frac{(1\times 10^{-3}kg)(10m/s^2)(10\Omega)}{(0.1m)(10^3\times 10^{-4}T)} = 10V$$

119. (*b*) Force between two parallel current carrying wires

$F \propto I_1I_2$

$\Rightarrow F_1 : F_{21} = 1:4$

120. [2] For equilibrium, Gravitational force acting on the wire = Magnetic force on the wire

$Mg = IlB$

$$I = \frac{mg}{\ell B} = \frac{40\times 10^{-3}\times 10}{50\times 10^{-2}\times 0.4} = 2A$$

121. (*b*) Net force of charge,

$\vec{F}_{net} = q\vec{E} + q(\vec{v}\times\vec{B})$

$= q\vec{E} + q|\vec{v}||\vec{B}|\sin 90°$

$= (2q\hat{i} + 3q\hat{j} + 0)$

Work done W

$= \vec{F}_{net}\cdot\vec{S} = (2q\hat{i}+3q\hat{j})\cdot(\hat{i}+\hat{j})$

$= 2q + 3q = 5q$

122. (*d*) $U = \frac{B^2}{2\mu}V$, where $\mu = 2\mu_0$.

$U = \frac{B^2}{4\mu_0}V$, and $V = Al$.

Thus, $U = \frac{B^2Al}{4\mu_0}$.

123. (*a*) Voltage sensitivity $= \frac{\theta}{V} = \frac{NAB}{CR}$

$$\text{Ratio} = \frac{(N_1A_1B_1)/R_1}{(N_2A_2B_2)/R_2}$$

$$= \frac{15\times 3.6\times 0.25}{21\times 1.8\times 0.50}\times\frac{7}{5} = 1$$

124. (*d*)

(A) From $\tau = C\theta$, the torsional constant has the dimensional formula:

$[C] = [ML^2T^{-2}]$

(B) Current Sensitivity (C.S)

$$C.S = \frac{\theta}{I} = \frac{BNA}{C} \quad (\because C\theta = iANB)$$

So, N, hence C.S.

Voltage Sensitivity:

$$V.S = \frac{\theta}{V} = \frac{BNA}{RC}$$

Since R depends on N, increasing current sensitivity does not always increase voltage sensitivity.

(C) Voltage Sensitivity (V.S) Dependency

$$V.S \propto \frac{NAB}{CR}$$

If R also increases with N, voltage sensitivity does not necessarily double.

(D) Incorrect Theory

MCG to ammeter conversion requires low shunt resistance, not large.

(E) Current Sensitivity Dependency

$C.S \propto N$

Since C.S is directly proportional to N, the statement claiming an inverse relation is incorrect.

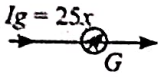
125. (*c*) $I \propto \theta$ (angle of deflection)

$$\therefore \frac{I_2}{I_1} = \frac{\theta_2}{\theta_1} \Rightarrow \frac{I_2}{200\mu A} = \frac{\pi/10}{\pi/3} = \frac{3}{10}$$

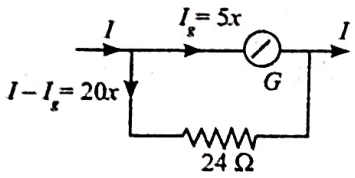
$\Rightarrow I_2 = 60\ \mu$A

126. (*b*) Given shunt = 24 Ω

Let x = current/division



After applying shunt deflection in moving coil galvanometer falls from 25 divisions to 5 division

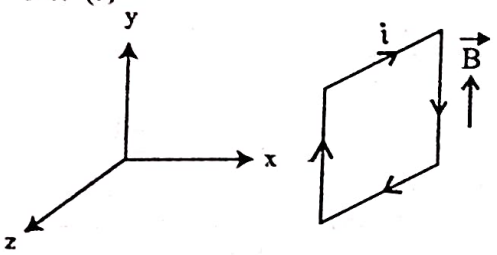


Now $5x \times G = 20x \times 24$

$G = 4 \times 24$

$G = 96\ \Omega$

127. (*b*)



Magnetic moment, $\vec{M} = i\vec{A}$

$= 5 \times (0.2) \times (0.1) \times (-\hat{i})$

$= 0.1\,(-\hat{i})$

Torque,

$\vec{\tau} = \vec{M} \times \vec{B} = 0.1(-\hat{i}) \times \left(2 \times 10^{-3}\right)(\hat{j})$

$= 2 \times 10^{-4}\ (-\hat{k})$ N – m

Direction of torque of loop is $(-\hat{k})$

128. [4] $\tau = BINA\sin\phi$

$C\theta = \text{BIN}A\sin 90°$ [$\therefore \tau = C\theta$]

$$\Rightarrow C = \frac{\text{BINA}}{\theta}$$

$$= \frac{0.01 \times 10 \times 10^{-3} \times 100 \times 2 \times 10^{-4}}{0.05}$$

$= 4 \times 10^{-5}$ N-m/rad.

$\therefore x = 4$

129. [5] The current, $i = \dfrac{V}{R}$

$i = K\theta$

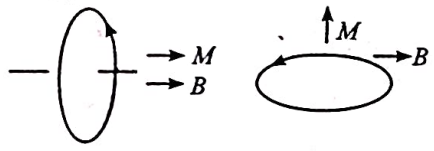
$$\frac{2}{G+R} = K\theta$$

$$\Rightarrow \frac{1}{\theta} = \frac{(G+R)K}{2} = R\left(\frac{K}{2}\right) + \frac{KG}{2}$$

$$\text{Slope} = \frac{K}{2} = \frac{1}{4}$$

$\Rightarrow K = 0.5 = 5 \times 10^{-1}$ A

130. [2]



initial $W_2 = \Delta U$ final

$\because (\Delta U = -\vec{M} \cdot \vec{B})$

$\left(-\vec{M} \cdot \vec{B}\right)_f - \left(-\vec{M} \cdot \vec{B}\right)_i$

$= -\text{MB} \cos 90° + \text{MB} \cos 0°$

$= \text{MB} = \text{NIAB}$

$= 200 \times 100 \times 1^{-6} \times \frac{5}{2} \times 10^{-4} \times 1 = 5\,\mu\text{J}$

131. [100] $W = \Delta U = U_2 - U_1$

$W = (-\vec{\mu}.\vec{B})_2 - (-\vec{\mu}.\vec{B})_1$

$= 0 + (\vec{\mu}.\vec{B})_1 = 100\text{gJ}$

$= (100 \cdot 5 \cdot 10^{-3} \cdot 1 \cdot 10^{-3}) \cdot 0.2$ J

$= 1 \cdot 10^{-4}$ J $= 100\mu$J

132. (*b*) The eddy currents are setup due to motion of coil which oppose the motion of coil thus bringing the coil to rest.

133. (*a*) Current sensitivity is the deflection (ϕ) per unit current (I) i.e:, $\dfrac{\phi}{I} = \dfrac{NAB}{K}$

Voltage sensitivity,

$$\frac{\phi}{V} = \frac{\phi}{IR_g} = \frac{\text{current sensitivity}}{R_g}$$

Voltage sensitivity $\propto$ current sensitivity (when R_g is constant)

So percentage change in V_s is 25%

134. (*d*) Current sensitivity $= \dfrac{BNA}{k}$

Current sensitivity $\propto N$

As N is doubled then current sensitivity is doubled.

$$\text{Voltage sensitivity} = \frac{BNA}{kR}$$

$$\text{Voltage sensitivity} \propto \frac{N}{R}$$

As N is doubled R is also doubled.

So, there is no change in voltage sensitivity.